

Ex No: 3b

IMPLEMENTATION OF A FUZZY INFERENCE SYSTEM

Aim:

To implement Fuzzy Inference System.

Procedure:

Define the input range for temperature (0 to 40) and output range for fan_speed (0 to 10) using NumPy arrays.

Create triangular membership functions (trimf) for low, medium, and high categories for both temperature and fan speed.

Plot and display the membership functions using Matplotlib subplots.

Perform fuzzification on a crisp temperature input (28°C) using interp_membership to compute membership degrees.

Apply fuzzy rules matching temperature membership levels directly to fan speed levels.

Compute the crisp fan speed output using centroid defuzzification (defuzz) and display the final result.

Program:

Example: Simple Fuzzy Inference System for Temperature Control

```
import numpy as np
import skfuzzy as fuzz
import matplotlib.pyplot as plt

# Step 1: Define the fuzzy variables (input: temperature, output: fan speed)
temperature = np.arange(0, 41, 1) # Temperature from 0 to 40 degrees Celsius
fan_speed = np.arange(0, 11, 1) # Fan speed from 0 to 10

# Step 2: Define fuzzy membership functions
# Temperature membership functions: Low, Medium, High
temp_low = fuzz.trimf(temperature, [0, 0, 20])
temp_medium = fuzz.trimf(temperature, [10, 20, 30])
```

```
temp_high = fuzz.trimf(temperature, [20, 30, 40])

# Fan speed membership functions: Low, Medium, High
fan_low = fuzz.trimf(fan_speed, [0, 0, 5])
fan_medium = fuzz.trimf(fan_speed, [2, 5, 8])
fan_high = fuzz.trimf(fan_speed, [5, 10, 10])

# Step 3: Visualize the membership functions
plt.figure(figsize=(10, 6))

plt.subplot(2, 1, 1)
plt.plot(temperature, temp_low, label='Low')
plt.plot(temperature, temp_medium, label='Medium')
plt.plot(temperature, temp_high, label='High')
plt.title("Temperature Membership Functions")
plt.xlabel("Temperature (°C)")
plt.ylabel("Membership Degree")
plt.legend()

plt.subplot(2, 1, 2)
plt.plot(fan_speed, fan_low, label='Low')
plt.plot(fan_speed, fan_medium, label='Medium')
plt.plot(fan_speed, fan_high, label='High')
plt.title("Fan Speed Membership Functions")
plt.xlabel("Fan Speed")
plt.ylabel("Membership Degree")
plt.legend()

plt.tight_layout()
plt.show()
```

```

# Step 4: Fuzzification - Define a crisp temperature input
temperature_input = 28 # Example temperature

# Fuzzify the input
temp_low_level = fuzz.interp_membership(temperature, temp_low, temperature_input)
temp_medium_level = fuzz.interp_membership(temperature, temp_medium,
temperature_input)
temp_high_level = fuzz.interp_membership(temperature, temp_high, temperature_input)

# Step 5: Apply fuzzy rules
# Rule 1: If temperature is Low, fan speed is Low
# Rule 2: If temperature is Medium, fan speed is Medium
# Rule 3: If temperature is High, fan speed is High

fan_low_level = temp_low_level
fan_medium_level = temp_medium_level
fan_high_level = temp_high_level

# Step 6: Defuzzification - Calculate the crisp output (fan speed)
fan_output = fuzz.defuzz(fan_speed, fan_low * fan_low_level + fan_medium *
fan_medium_level + fan_high * fan_high_level, 'centroid')

# Step 7: Print the output
print(f"Temperature: {temperature_input}°C")
print(f"Fuzzified fan speed: {fan_output:.2f}")

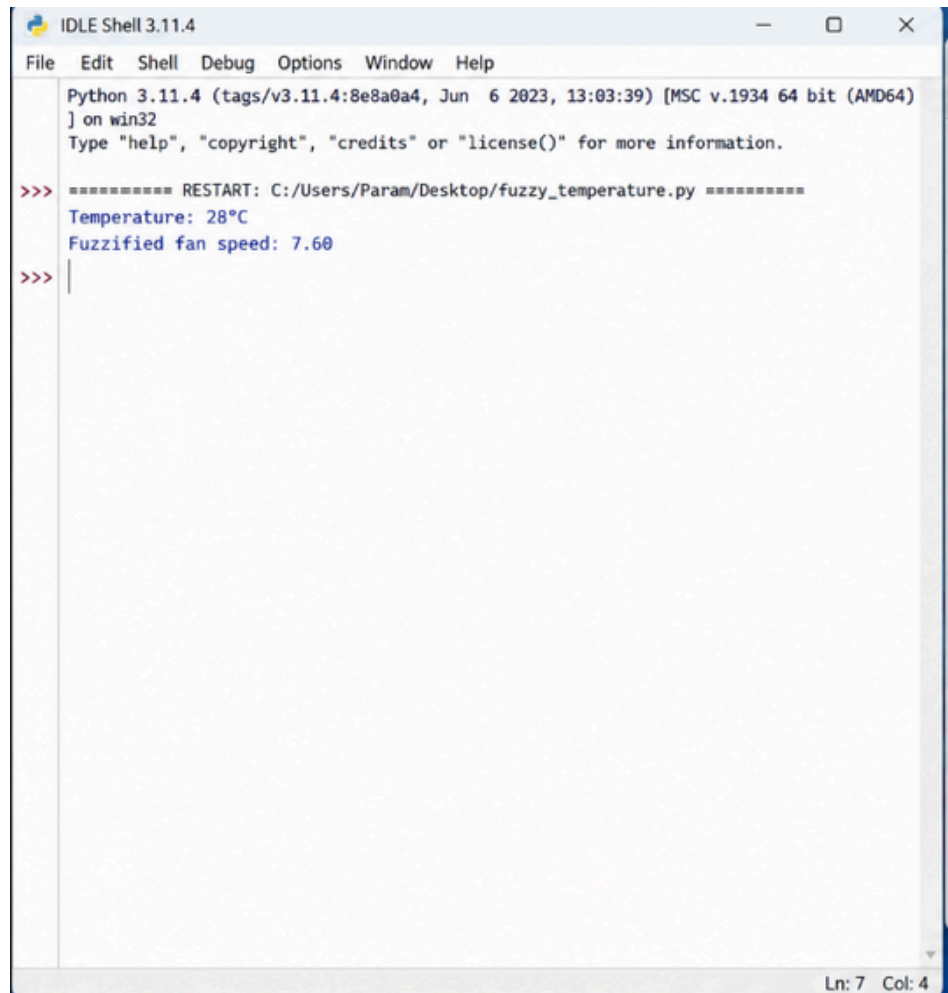
```

Output

```

Temperature: 28°C
Fuzzified fan speed: 6.00

```



The image shows a screenshot of the IDLE Shell 3.11.4 window. The window has a title bar with the text "IDLE Shell 3.11.4" and standard window controls (minimize, maximize, close). Below the title bar is a menu bar with the following items: File, Edit, Shell, Debug, Options, Window, and Help. The main text area contains the following text:

```
Python 3.11.4 (tags/v3.11.4:8e8a0a4, Jun 6 2023, 13:03:39) [MSC v.1934 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.

>>> ===== RESTART: C:/Users/Param/Desktop/fuzzy_temperature.py =====
>>> Temperature: 28°C
>>> Fuzzified fan speed: 7.60
>>> |
```

The status bar at the bottom right of the window displays "Ln: 7 Col: 4".